

[20]

SARDAR PATEL UNIVERSITY
B.Sc (CA & IT) (SEMESTER-VI) EXAMINATION July-2021
27 July, 2021, Monday
02:00 p.m. to 04:00 p.m.
PS06CIIT27 (Information Security)

Marks: 70
[10]

Q. 1 Multiple Choice Questions

- 1 Which action is that compromises the security of information owned by an organization?
A. Threat B. Security Attack C. Service Attack D. None of these
- 2 The ____ attacks are in nature of eavesdropping on, or monitoring of transmission.
A. Active Attack B. Passive Attack C. Service Attack D. Both A & B
- 3 ____ is one type of security service.
A. Data Integrity B. Authentication C. Access Control D. Non-repudiation
- 4 cryptography is also known as ____
A. plain text writing B. Stream writing C. Secret writing D. None of above
- 5 The process of attempting to discover the plaintext or key is known as ____
A. Cryptanalysis B. Cipher text C. Stream cipher D. Block cipher
- 6 ____ is a mechanism that bypasses a normal security check.
A. Logic bomb B. Trojan horse C. Bacteria D. Trapdoor
- 7 When ____ are executed, it may produce one or more copies of itself to be activated later on the same or some other system.
A. Logic bomb B. Back doors C. Worms D. boot programs
- 8 What is firewall in computer network?
A. The physical boundary of network
B. An operating system of computer network
C. A system designed to prevent unauthorized access
D. A web browsing software
- 9 The ____ server in the DNZ needs only list a limited number of public IP addresses.
A. DNS B. NAT C. Proxy D. Firewall
- 10 ____ server is a server that creates a secure tunnel connection.
A. DNS B. VPN C. Tunnel D. authentication

Q. 2 Fill in the Blanks and True or False.**[08]**

- 1 Service threats exploit service flaws in computers to inhibit use by legitimate users. (True/False)
- 2 The ____ control is the ability to limit and control the access to host system and applications via communication links.
- 3 The exact substitutions and transformations performed by the algorithm depend on the plaintext key. (True/False)
- 4 ____ is the scrambled message produced as output.
- 5 What the virus does besides spreading is known as ____.
- 6 Infection mechanism also referred as Infection vector. (True/False)
- 7 ____ Controls access to service according to which user is attempt.
- 8 Behavior controls how particular services are used. (True/False)

Q. 3 Answer the Following.(Any 10)

[20]

- 1 Define threat & attack.
- 2 List the different types of security services.
- 3 Draw diagram of Model for network security.
- 4 What is cryptography? Give name of its component?
- 5 Define plain text and cipher text
- 6 Differentiate public key and private key.
- 7 Define: Virus and Worm
- 8 What do you mean by misfeasor?
- 9 Give two examples software threat that do not replicate.
- 10 Define Digital signature.
- 11 List types of firewall.
- 12 What is circuit level gateway?

Q.4 Answer the Following.(Any 4)

[32]

- 1 What is security attack? Explain in detail with its types?
- 2 List and briefly define categories of active and passive security attacks.
- 3 What are the types of traditional ciphers? Explain.
- 4 Explain the conventional /symmetric encryption principles and model of symmetric encryption.
- 5 Explain different nature of viruses in detail
- 6 Write a brief note on Intruders.
- 7 List and explain four general techniques of firewall use to control access and enforce the security policy.
- 8 What is the difference between packet filtering and stateful inspection firewall.

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